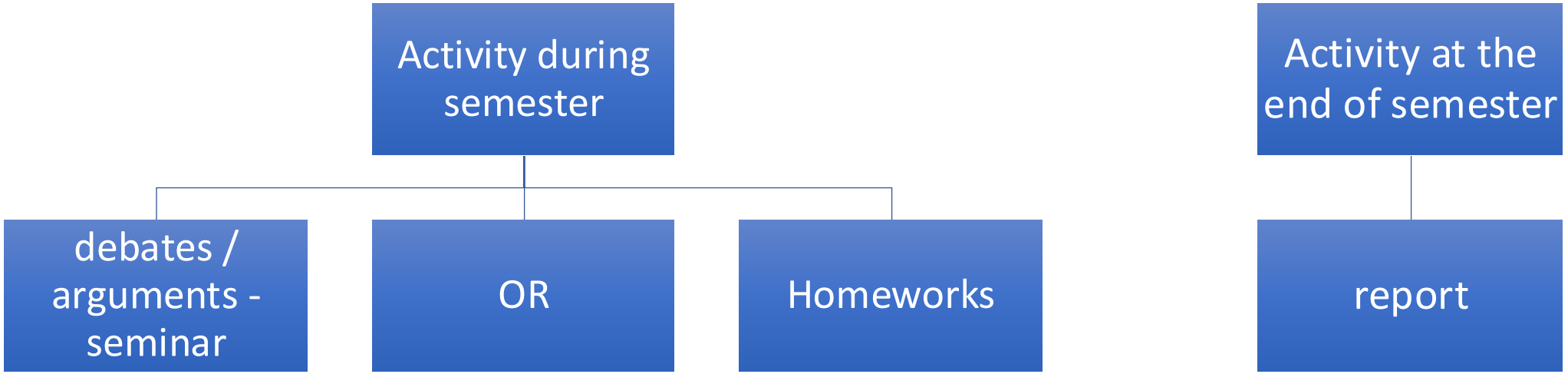
Academic Ethics and Integrity in Computer Science / Data Science

Organization

- Course – 2 h / week – course directory on Teams: 3vxoeoz
- Seminar 2h/every two weeks
- **Debates/HW + report**



Points

- **10 points** – seminar activity

  OR

- **7 points** – HW
- **30 points** – report

- BONUS: Course responses

A little bit of history

- ACM Code of ethics (professional) – 1973
- Walter Maner – starter kit – ”Teaching Computer Ethics” - 1978
- Moor,J, "What Is Computer Ethics",*Metaphilosophy*,Vol.16,No.4, Oct 1985, pp. 266-275
- Johnson, Deborah, G, *Computer Ethics*, Prentice-Hall, 1985
- Terrell Ward Bynum – 40 univ. in UK
  - + Simon Rogerson – first research center in CS ethics - De Monfort Univ.
- Ethics4EU project

What?

- Are there any ethical aspects regarding IT domain – which are they?
- How can I recognize these aspects?
- Detect, analyze, solve the ethical dilemma

- Codes:

1. ACM https://www.acm.org/about-acm/acm-code-of-ethics-and-professional-conduct
2. IEEE-CS/ACM - Software Engineering https://www.ieee.org/about/corporate/governance/p7-8.html

ACM Code of Ethics and Professional Conduct (updated 2018)

1. GENERAL ETHICAL PRINCIPLES: *A computing professional should...*

1.1 Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing

1.2 Avoid harm

1.3 Be honest and trustworthy

1.4 Be fair and take action not to discriminate

1.5 Respect the work required to produce new ideas, inventions, creative works, and computing artifacts

1.6 Respect privacy

1.7 Honor confidentiality

ACM Code of Ethics and Professional Conduct

2. PROFESSIONAL RESPONSIBILITIES: *A computing professional should...*

2.1 Strive to achieve high quality in both the processes and products of professional work

2.2 Maintain high standards of professional competence, conduct, and ethical practice

2.3 Know and respect existing rules pertaining to professional work

2.4 Accept and provide appropriate professional review

2.5 Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks

2.6 Perform work only in areas of competence

2.7 Foster public awareness and understanding of computing, related technologies, and their consequences

2.8 Access computing and communication resources only when authorized or when compelled by the public good.

2.9 Design and implement systems that are robustly and usably secure

ACM Code of Ethics and Professional Conduct

3. PROFESSIONAL LEADERSHIP PRINCIPLES: *A computing professional, especially one acting as a leader, should...*

3.1 Ensure that the public good is the central concern during all professional computing work

3.2 Articulate, encourage acceptance of, and evaluate fulfillment of social responsibilities by members of the organization or group

3.3 Manage personnel and resources to enhance the quality of working life

3.4 Articulate, apply, and support policies and processes that reflect the principles of the Code

3.5 Create opportunities for members of the organization or group to grow as professionals

3.6 Use care when modifying or retiring systems

3.7 Recognize and take special care of systems that become integrated into the infrastructure of society

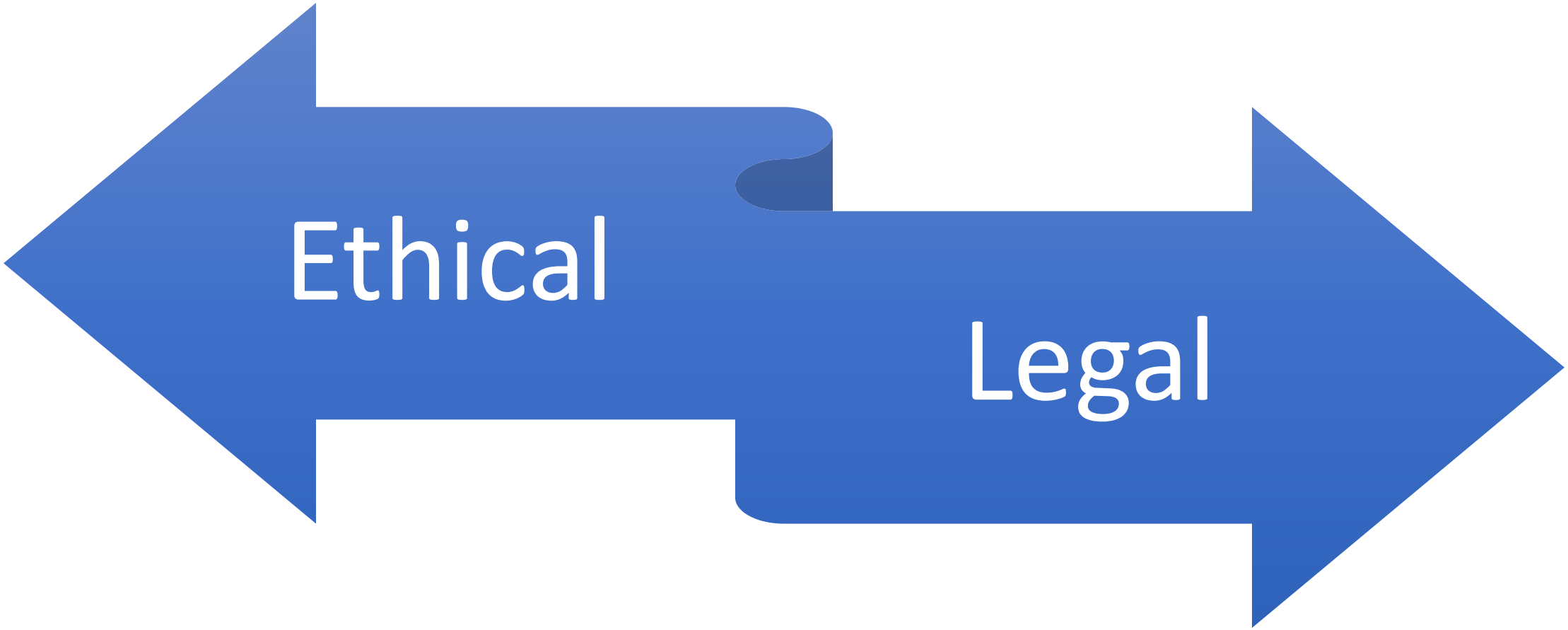
ACM Code of Ethics and Professional Conduct

4. COMPLIANCE WITH THE CODE: *A computing professional should...*

4.1 Uphold, promote, and respect the principles of the Code

4.2 Treat violations of the Code as inconsistent with membership in the ACM

Quiz 1:



Ethics

- As a guideline to users
- Free to follow
- Universal
- Immoral



Credits: eckerson.com

Law

- As a rule to users
- Must be followed
- Depend on legislation (contry, EU)
- Fellony, crime, infrigiment



Credits: Hiclipart.com

Quiz 2

- Lack of ethics in IT can lead to …

Examples

Security/ safety

Fraud

Harm Environment protection

Conflict of interests

Theft

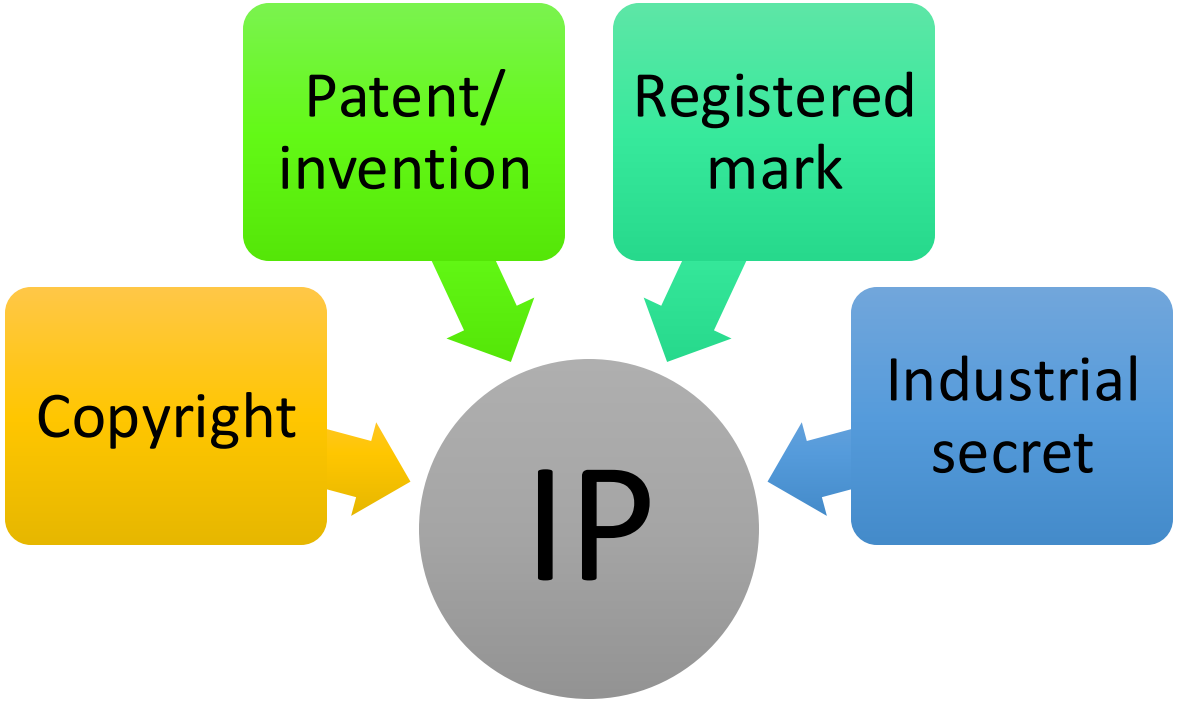
Plagiarism

Health

Intellectual Property (IP)

Intellectual property

- Refers to creations: ideas, inventions, technologies, artistic works, symbols, names, images, designs
- ***= set of legal rights that represent the outcome of intellectual activities from industry, science and artistic domains***
- May belong to a group or individual



Copyright

- Statute of Anne /Queen Anne, England, ≈ 1710
- USA, 1790
- France, 1793

Did you know:

Law 8 / 1996:

- Art. 1. - (1) Dreptul de autor asupra unei opere literare, artistice sau stiintifice, precum si asupra oricaror asemenea opere de creatie intelectuala, este recunoscut si garantat in conditiile prezentei legi. Acest drept este legat de persoana autorului si comporta atribute de ordin moral si patrimonial.

(Copyright of a literary, artistic of scientific work, as well as of any such work of intellectual creation, is recognized and guaranteed in the conditions of the present law. This right is related to the person of the author and inplies moral and patrimony attributes)

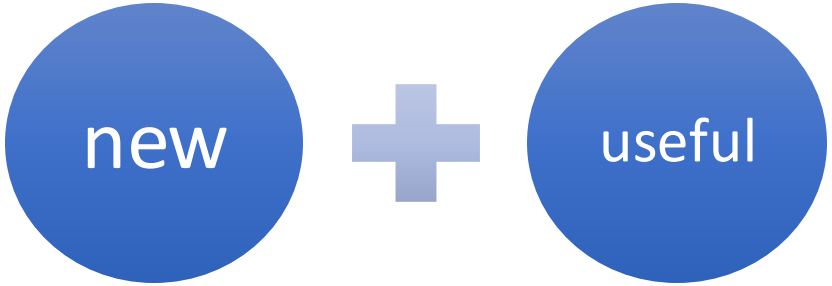
- (2) Opera de creatie intelectuala este recunoscuta si protejata, independent de aducerea la cunostinta publica, prin simplul fapt al realizarii ei, chiar neterminata.

(The work of intellectual creation is recognized and protected, independent of bringing it to the public knowledge, through the simple fact of obtain it, even unfinished)

Copyright – cont.

- Depend on contry
  - Ex. USA: 3 rules: originality, establish, expression
- Duration:
  - In Romania: lifetime and it is inherited
  - In USA: lifetime and 50 years after
- May generate revenues

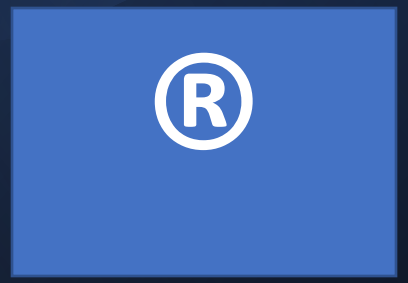
Patent



- Invention

- www.osim.ro - **OFICIUL DE STAT PENTRU INVENŢII ŞI MĂRCI**
- Grants rights to an inventor for a fixed period
- Obtained based on a standardized procedure
- Depends on the country
- International Patent

Trademark(Marcă înregistrată / comercială)



Is a form of intellectual property

Insignia (name / logo / symbol / image)

Used by a person or a legal entity (company, association a.s.o.)

Duration = 10 years

Community mark –EU – 10 years

Trade secret (Secret industrial)

- Formula, process, design, device etc.
- Offers the owner a competitive advantage
- Characteristics:
  - Secret information within organization
  - Plan for keeping the secret – NDA = Non Disclosure Agreement
  - Value
  - Effort and money invested for obtaining it



Software

- Legea 8/1996, art.72, 2) “Ideile, procedeele, metodele de functionare, conceptele matematice si principiile care stau la baza oricarui element dintr-un program pentru calculator, inclusiv acelea care stau la baza interfetelor sale, nu sunt protejate.” – drepturi de autor

(Ideas, procedures, functioning methods, mathematical concepts and principles that form the basis of a computer program, including those for its interfaces, are not protected – by copyright)

- Software – can be protected by:

❑ Copyright:
  - includes source code, object code, executable code, ma
  - Use
  - Must be obtained: ORDA, RNPC
  - EU framework for legal protection of software: here

❑ Patent: hardware components, processes, ideas i
  - Ex. https://www.google.com/patents/US9292694

❑ Trade mark

❑ Trade secret - NDA

Who: physical or legal person

What: activities: software development/import/distribution/rent /comercialization

[https://www.avocatoo.ro/blog/cum-inregistrezi-un-software-la-orda]

Copyright

Included:

1. Make copies
2. Distribute copies
3. Display or perform the work in public
4. Make any derivate work

What is not covered?:

- Ideas
- Facts
- Discoveries

Limitations:
- Time
- Fair use

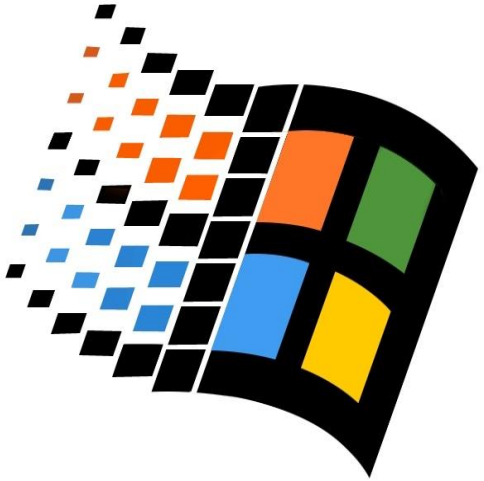
Cases



1989-1992: Apple vs. Microsoft

- În 1989, Apple accused Microsoft and HP for violation of copyright for the “look & feel” interface
- Apple lost the trial in 1992

1998 – 2001 US vs. Microsoft



- Antitrust law

- Microsoft was accused of monopoly power and anti-competitive practices



Marvel vs. onlineGame (2006)





- NCSoft was sued by Marvel for the following reason:
- The game XX produced by NCSoft allowed users to create avatars based on any character, for example Spiderman a.s.o.
- Is Marvel entitled to ask for copyright?

- “companies settled …, the terms weren't disclosed.”

Debate cases

Case 1 (copyright OEC – Online Ethics Center, http://www.onlineethics.org)

1. Company A organizes an idea contest / hackaton, with the promise to recognize, award and support the employees with their business ideas, even to open their own company
2. 100 participants => 5 selected projects
3. Employee Z with project PZ – continues alone, in his free time to develop the product, without any support from company A
4. After 3 months, he has an MVP (Minimum Viable Product), when divison X from the company A grants him financial support for a market research. Then, decides not to develop a new company but to further develop the PZ as an internal product.
5. Work on documentation to obtain a patent: inventor Z, owner X. (Z remembers that at hiring it signed a standard form for inventions / patents/ copyrights)
6. After 6 months, division X decides to move product PZ in another division Y, in another city, and Z has no role in it

- Q1: What can Z do? What he should have than differently?
- Q2: What should the company have than differently (ethically)?

Case 2 (copyright OEC – Online Ethics Center, http://www.onlineethics.org)

- In1994, Lem El-Zif, researcher at UI (Univ. of Illinois) discovered a new compression method for images and text for web pages
- UI obtains a patent and assigns to firm ADG for managing revenues
- Terry is a student at UI and he is helping his highschool with a web page and writes a program webCompress that incorporates Lem El-Zif algorithm. Terry loads his program on the site and states that it can be used freely.
- The highschool receives a letter that asks for a payment for using the patent of Lem El-Zif algorithm

Q1: What are the ethical issues in this case?

Q2: What might each participant want from this situation?

Q3: What kinds of solutions might each participant find acceptable or unacceptable?

Case 3 (copyright FNO http://www.fno.org/jun96/legal.html)

Surfing Photographs

- A student is preparing her project for English classes that should contain images. She finds a nice surfing picture on a website of a beach resort.

Q1: Is it legal to use the image taking into consideration the “fair use” stated in its copyright?

- Impressed by the project, the teacher proposes to publish the paper on the school’s website.

Q2: Can the image be used? On what terms?